

OLNEY-NOBLE, IL, USA

WMO: 744659

Lat: 38.722N

Lon: 88.176W

Elev: 482

StdP: 14.44

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53822

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	10.2	-7.7	3.7	11.9	-1.3	5.2	15.5	26.1	31.9	23.6	29.3	5.1	300	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	94.6	76.3	91.2	75.4	89.9	75.1	80.7	88.8	78.9	86.9	77.5	85.8	5.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	152.9	85.0	76.8	142.2	83.0	75.0	133.5	82.0	44.8	88.9	42.8	87.4	41.3	84.9	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.3	17.8	15.8	-1.9	98.2	6.0	7.8	-6.2	103.8	-9.8	108.4	-13.2	112.8	-17.5	118.5		
			-2.7	83.2	5.9	2.8	-6.9	85.2	-10.4	86.9	-13.6	88.4	-17.9	90.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.7	30.6	34.0	45.5	55.9	66.1	74.6	76.8	75.6	69.7	57.3	44.9	35.6
	DBStd	18.69	11.52	11.40	11.09	9.62	8.32	5.74	5.55	5.36	6.96	9.19	9.71	10.00
	HDD50	2039	608	460	222	50	2	0	0	0	0	30	210	457
	HDD65	4763	1066	868	609	296	93	4	1	1	33	276	604	912
	CDD50	4107	7	12	82	228	500	739	832	794	590	256	57	10
	CDD65	1358	0	0	5	24	126	293	367	331	173	38	1	0
	CDH74	13143	0	0	32	222	1107	2817	3686	3142	1736	389	12	0
	CDH80	5250	0	0	3	35	327	1128	1608	1339	693	116	1	0
Wind	WSAvg	5.9	7.8	7.5	7.7	7.6	5.9	4.7	3.6	3.4	4.0	5.5	6.6	7.1
Precipitation	PrecAvg	43.8	2.8	2.6	4.3	4.3	4.8	4.1	4.0	3.5	2.8	3.4	4.2	3.5
	PrecMax	58.0	8.7	5.9	9.1	9.6	12.3	8.4	9.6	11.2	6.7	9.8	12.2	10.0
	PrecMin	27.9	0.1	0.5	0.6	1.5	0.9	0.5	0.6	0.5	0.0	0.6	0.2	0.7
	PrecStd	8.3	2.1	1.4	2.1	2.0	2.9	2.3	2.2	2.2	1.8	2.1	2.2	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.5	68.0	80.5	83.8	90.8	97.0	102.0	99.3	93.4	88.5	74.9	64.2
		MCWB	59.4	60.9	64.2	66.6	71.4	77.1	75.5	77.9	73.7	72.0	60.9	57.9
	2%	DB	58.6	62.8	72.6	80.7	86.5	91.4	94.2	93.3	90.7	82.4	70.0	59.4
		MCWB	54.1	56.1	61.4	64.5	70.8	75.6	77.4	75.7	72.7	68.1	61.5	55.6
	5%	DB	54.0	56.8	67.9	75.4	83.7	90.0	90.9	90.6	88.0	78.7	64.2	54.8
		MCWB	50.1	50.3	58.8	63.1	69.6	74.5	77.1	75.6	71.7	65.3	56.8	50.6
	10%	DB	46.5	51.8	63.4	72.5	81.1	87.5	88.3	87.9	83.8	72.9	61.0	50.4
		MCWB	42.9	46.7	56.6	61.0	68.4	73.8	76.4	74.8	69.8	62.9	53.5	46.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	62.7	65.9	71.0	76.5	81.4	83.2	83.0	78.3	74.8	66.2	61.1
		MCDB	63.0	65.7	75.1	80.2	83.2	88.6	90.5	92.3	88.2	83.9	71.9	62.9
	2%	WB	55.4	57.7	63.3	67.4	74.0	78.7	81.1	80.1	76.3	71.4	62.8	55.9
		MCDB	57.5	61.7	70.4	75.9	81.8	87.6	88.3	88.7	85.0	79.0	66.2	58.2
	5%	WB	50.6	52.0	60.5	65.1	72.2	77.1	79.4	78.2	74.2	68.5	58.7	51.9
		MCDB	53.0	56.0	66.3	71.8	79.8	85.8	87.0	86.4	82.6	75.6	63.9	54.8
	10%	WB	43.5	45.9	57.6	62.7	70.3	75.4	77.9	76.6	72.2	64.8	54.1	46.4
		MCDB	46.8	50.6	63.2	69.8	77.6	83.3	85.6	84.3	80.4	71.3	59.8	49.1
Mean Daily Temperature Range	5% DB	MDBR	15.0	16.1	18.2	20.4	19.4	19.9	19.9	21.5	23.4	21.5	17.9	14.1
		MCDBR	20.8	23.2	23.4	24.5	22.7	23.2	22.8	26.0	27.0	26.7	23.6	19.2
		MCWBR	18.2	18.5	14.5	13.4	9.9	9.9	9.3	12.0	11.7	14.7	16.3	16.1
	5% WB	MCDBR	19.2	21.4	21.0	21.2	19.5	19.2	19.0	21.6	23.5	22.8	20.7	17.1
MCWBR		18.4	19.0	14.7	14.1	10.9	11.1	11.1	12.4	12.8	15.1	16.5	16.0	
Clear-Sky Solar Irradiance	taub	0.310	0.322	0.356	0.404	0.439	0.441	0.464	0.467	0.414	0.360	0.319	0.307	
	taud	2.450	2.400	2.374	2.241	2.196	2.216	2.154	2.134	2.275	2.421	2.485	2.468	
	Ebn at Noon	275	286	286	277	268	267	259	255	263	269	269	267	
	Edh at Noon	26	31	35	43	46	45	48	47	39	30	25	24	
All-Sky Solar Radiation	RadAvg	609	858	1177	1550	1746	2020	1959	1779	1482	1067	718	519	
	RadStd	63	92	108	90	152	130	124	101	104	111	79	51	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	+140

Nomenclature: See separate page